



IBM Quantum



OFFICIAL EVENT

Qiskit

Fall Fest

2026

Foundations — Qubits, Gates & Superposition



► Speaker Introduction

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Dustin Rippe is a Qiskit Advocate and an IBM Certified Developer, holding both the Quantum Computation using Qiskit v2.X and v0.2X certifications.

His original background is in computer security (MS in Information Security, CISSP, etc), and he also holds a BS in Mathematics and Physics.



Topics

What We Will Cover



- **Qiskit Setup**

- Google Colab, VS Code, API Key and CRN

- **Quantum Computing**

- What Does Quantum Mean?, Qubits, States and Evolution

- **Vectors and Matrices**

- Vectors, States as Vectors, Qiskit's Statevector, Inner Products, Matrices, Matrices Act on Vectors, Qiskit's Operator, Combining Operations

Qiskit Setup

Preparing Your Environment



Qiskit is a Python-based software system for quantum computing.

It can be run in almost any Python environment, but normally Jupyter notebooks running in Google Colab, VS Code, or similar environment are used.

Google Colab is *browser-based* and free for anyone with a Google account.

VS Code runs *locally on your computer* and is also free to download, but requires additional setup.

For ease of use, I recommend starting in Google Colab.

Qiskit Setup

Google Colab



Sign-in to Google and go to: <https://colab.research.google.com/>

To run a Jupyter notebook in Colab:

- Click ***Upload Notebook***.
- Click ***Browse*** or drag-and-drop the notebook onto the dialog box.
- Once it uploads it will automatically open.

Notebooks can also be uploaded to Google Drive and launched from there.

Qiskit Setup

VS Code



VS Code is downloaded from: <https://code.visualstudio.com/download>

VS Code can be used for many programming languages, so Python and Jupyter specific extensions are required:

Jupyter by Microsoft

<https://marketplace.visualstudio.com/items?itemName=ms-toolsai.jupyter>

Python by Microsoft

<https://marketplace.visualstudio.com/items?itemName=ms-python.python>

Follow the *Install the Qiskit SDK* steps in IBM's guide:

<https://quantum.cloud.ibm.com/docs/en/guides/install-qiskit>

Qiskit Setup

API Key and CRN



If you do not have an IBM Cloud account already, make a trial account:
https://cloud.ibm.com/registration?utm_content=quantum-trial

Login at: <https://quantum.cloud.ibm.com/signin>

There is a small section labeled **API key**, click **Create+**, then enter a name for the key and click **Create**.

The key will be displayed for 5 minutes. Download or copy and save it.

In the **Instances** section of the dashboard click **View all**, click **Create instance+**, enter the details and copy the **CRN**.

Quantum Computing

What Does Quantum Mean?



Quantum mechanics is the theory used to describe the behavior of physical systems at very small scales.

Quantum computing uses these behaviors to perform calculations.

We will not be covering all of quantum mechanics, but there are some important concepts that we will use:

- Physical systems have a ***state***.
- Operations can change, or ***evolve***, the state.
- Measurements of the system are ***probabilistic*** in nature.

Quantum Computing

The Qubit



Classical computers use *bits* when they calculate.

Quantum computers use quantum bits, called *qubits*, for their calculations.

A qubit is a physical system that can be controlled as a *two-state quantum system*.

Qubits can be made from many kinds of quantum systems: electron or nuclear spin, energy levels in an atom or ion, photon polarization, or energy levels in a superconducting circuit.

Quantum Computing

States and Evolution



A quantum computation can be viewed as:

Initial State → *Operations* → *Final State*

Our goal is to use our knowledge of the rules of quantum state evolution to perform mathematical calculations that would otherwise require far too many resources to classically implement.

The way these quantum rules will be described is with **vectors** and **matrices**.

Vectors and Matrices

Vectors



Vectors are 1-dimensional arrays of numbers.

They can be **row vectors** or **column vectors**:

$$[1 \quad 2 \quad 3 \quad 4]$$

$$\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

Vector addition: $\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Scalar Multiplication: $k \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ kb \end{bmatrix}$

$$3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

Vectors and Matrices

States as Vectors



Qubits are often represented with a **column vector** called a **ket**. Each ket has an associated **row vector** called a **bra**.

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad \text{and} \quad \langle\psi| = [\alpha^* \quad \beta^*]$$

α and β are **complex numbers**, and α^* and β^* are their **complex conjugates**.

Complex conjugation means that if $\alpha = x + iy$, then $\alpha^* = x - iy$.

Importantly, the state represented by α and β is **normalized**: $|\alpha|^2 + |\beta|^2 = 1$

Vectors and Matrices

Qiskit's Statevector Class



Qiskit provides a class that represents the quantum state vector:

```
from qiskit.quantum_info import Statevector
sv = Statevector([1, 0])
```

Useful methods:

```
.from_circuit()      .from_label()
.conjugate()  .evolve()  .is_valid()  .inner()  .tensor()
.expectation_value()  .probabilities_dict()  .sample_counts()
```

See IBM's documentation for full details:

https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.quantum_info.Statevector

Vectors and Matrices

Vectors Need a Basis



We always choose a **basis** to work in. The *standard* or *computational basis states* in quantum computing are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

A qubit in a **pure state** can be written in several equivalent ways:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

This **linear combination** of basis states is called **superposition**.

Vectors and Matrices

Inner Products



The *inner product* combines a row and a column vector to produce a scalar.

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = ac + bd \qquad \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = 11$$

For quantum states the inner product is written using a bra and a ket, $\langle \psi | \psi \rangle$:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \langle \psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

$$\langle \psi | \psi \rangle = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = |\alpha|^2 + |\beta|^2 = 1 \qquad \langle 0 | 1 \rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Vectors and Matrices

The Matrix



A matrix is a 2-dimensional array of numbers:

Matrix addition:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

Matrix-Vector product:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix}$$

Matrix-Matrix product:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$$

Vectors and Matrices

Matrices Act on States



Operators are represented by matrices. They are linear transformations that act on the vectors representing quantum states:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

If an operator A acts on $|\psi\rangle$, then: $A|\psi\rangle = |\psi'\rangle$

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Vectors and Matrices

Operator



Qiskit provides the **Operator** class and several other useful classes, including **SparsePauliOp**.

```
from qiskit.quantum_info import Operator
op = Operator([[0, 1], [1, 0]])
```

Useful methods:

```
.from_circuit()      .from_label()
.adjoint()   .conjugate()   .dot()   .is_unitary()   .tensor()
.power()     .to_instruction()   .to_matrix()
```

See IBM's documentation for full details:

https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.quantum_info.Operator

Vectors and Matrices

Combining Operations



Multiple operators can be applied in sequence.

$$|\psi\rangle \xrightarrow{A} |\psi'\rangle \xrightarrow{B} |\psi''\rangle$$

So we have $A|\psi\rangle = |\psi'\rangle$ and $B|\psi'\rangle = |\psi''\rangle$. We can write this as:

$$|\psi''\rangle = B(A|\psi\rangle) = BA|\psi\rangle$$

Since BA is regular matrix multiplication we can calculate it first to simplify.

$$BA = C, \quad |\psi''\rangle = BA|\psi\rangle = C|\psi\rangle$$

Vectors and Matrices

Two Types of Matrices

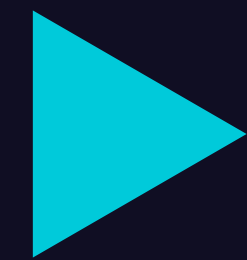


Unitary operators describe reversible evolutions of quantum states.

$$U^\dagger U = U U^\dagger = I$$

Hermitian operators describe **observables**, their real eigenvalues are the possible measurement outcomes for that observable.

$$A^\dagger = A$$



Questions?

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Qiskit Fall Fest 2026 at University of Engineering & Technology Lahore, Narowal Campus